

1. Suppose a star has a parallax $p = 0.01$ arcsec, peak wavelength $\lambda_{max} = 250$ nm and apparent magnitude $m = +5$. What is its (a) Distance in parsec? (b) Absolute magnitude M ? (c) Luminosity in $L_{\odot}$? [assume that the absolute magnitude of the Sun is $+5$] (d) Surface temperature? (e) Radius? (f) Angular radius (radian and arcsec)? (g) Solid angle (steradian)?

[8]

2. Assume the Earth's atmosphere to be a layer around the surface with a constant thickness. Define θ to be the angle of a celestial object from the zenith.

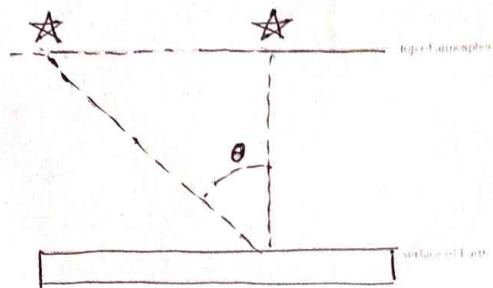


Figure 1: .

- Show that the optical depth is given by $\tau = \tau_0 \sec(\theta)$, where τ_0 is the optical depth of the atmosphere in the direction of the zenith ($\theta = 0$).
- Suppose I_0 is the intensity of the celestial object above the atmosphere. Find out how this intensity will change as a function of the angle θ .
- Show that the magnitude will vary as a function of θ as $m(\theta) = m(\theta = 0) + A \sec(\theta)$, where A is a constant.

[2+1+2]

3. Suppose a planetary nebula emits radiation with emissivity j (energy/time/volume/solid angle), within a spherical shell whose thickness ΔR is 10% of its radius R . Assume there is negligible absorption in the shell and that the centre of the shell is hollow and so it emits no radiation.

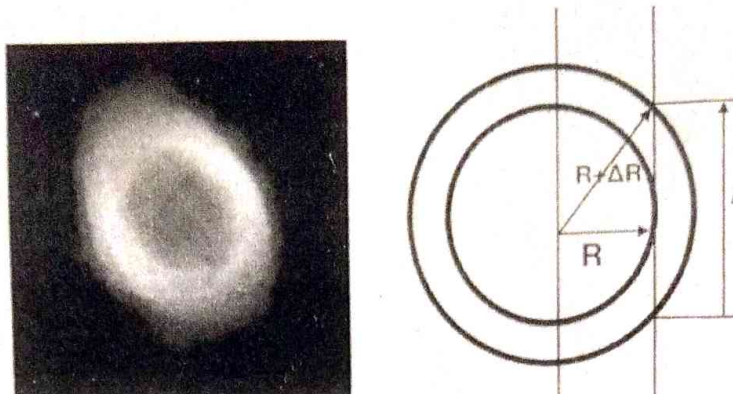


Figure 2: M57 (Ring Nebula) and a schematic diagram for Q3.

- Using the fact that the thickness $\Delta R \ll R$, one can derive an approximate expression for the total luminosity L of the shell in terms of emissivity j and R . Recall that the total luminosity is given by an integral over volume and solid angle of the emissivity j :

$$L = \int d\Omega \times 4\pi \int_R^{R+\Delta R} j r^2 dr.$$

Take the result of the angular integration to be 4π . Show that for $\Delta R \ll R$, one has

$$L \approx 16\pi^2 j R^2 \Delta R.$$

- Derive an expression for the length of the longest line of sight through the shell (marked in the diagram as the chord labelled l), and compare this with the length of the radial line through the shell. Show that while the length of the radial line through the centre is $2 \times \Delta R$, (because the radial line cuts through both the front and back of the shell), the maximum length of the chord is given by $l \approx 2\sqrt{2R\Delta R}$.
- Compute the ratio of intensity along this longest limb line to that along radial line. Show that the ratio is given by $I_{\text{limb}}/I_{\text{centre}} = \sqrt{2R/\Delta R}$. [Hint: use the expression for intensity in terms of emissivity for low optical depth.]
- Use this result to explain the 'ring-like' appearance of the planetary nebula. [Hint: For $\Delta R \ll R$, the ratio of intensity in the limb and in the centre is much larger than unity.]

[2+2+2+1]